Practice workbook · Estimate an unmeasured value by interpolating or extrapolating data

Data give values on either side of the target, or a steady trend that can be extended, and a value not measured is to be estimated.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-ESTIMATE-INTERPOLATE-EXTRAPOLATE

1. Question 4 · part b · 4 marks

June 2025 Paper 2

Worked-example retry · Attempt part b only.

4 A student investigates the effect of different concentrations of sugar solution on the mass of cylinders made from potato.

This is the student's method.

- push a cork borer into different parts of a potato
- remove five cylinders of potato
- wrap each cylinder in a piece of filter paper
- remove from the filter paper and find the mass of each cylinder
- place one cylinder into distilled water
- place each of the other cylinders into different concentrations of sugar solution
- remove the cylinders after two hours and wrap in filter paper
- remove from the filter paper and find the new mass

These are the student's results.

1.00 au = −7 %	1.50 au = −10 %	0.00 au (distilled water) = +8 %
	0.25 au = +4 %	0.50 au = −1 %

The student's results are given as the percentage change in mass of the cylinder and the concentration of sugar is given in arbitrary units (au).

(a) (i) Draw a table to display the results.

(4)

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P 7 8 7 6 9 A 0 8 2 0

1. Question 4 · part b · 4 marks

June 2025 Paper 2

Worked-example retry · Attempt part b only.

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(ii) Determine the median value for sugar concentration.

(1)

median value = ...

(iii) The original mass of the potato cylinder placed into the sugar solution, of concentration 0.25 au, was 7.0 g.

Calculate the mass of the cylinder after two hours.

Give your answer to two significant figures.

(3)

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mass of cylinder = ... g

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(b) Use the student's results to estimate the concentration of sugar solution that would produce no change in the mass of the potato cylinder.

You should include details of how you obtained your answer.

(4)

concentration of sugar solution = ... au

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P 7 8 7 6 9 A 0 9 2 0

1. Question 4 · part b · 4 marks

June 2025 Paper 2

Worked-example retry · Attempt part b only.

(c) (i) Explain why filter paper was wrapped around the potato cylinders at the beginning and end of the investigation.

(2)

(ii) The student kept the potato cylinders in the liquid for two hours.

State two other factors that should be kept constant during this investigation.

(2)

1 .....

.....

2 .....

.....

(Total for Question 4 = 16 marks)

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1. Question 4 · part b · 4 marks · Working and answer

Working and answer

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2. Question 4 · part b.i · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.i only.

4 (a) Lipids are digested in the small intestine.

(i) Describe a test for lipids in food.

(3)

(ii) Name the enzyme that digests lipids.

(1)

(iii) Name the products of lipid digestion.

(2)

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(b) A student investigates how the pH of a lipid solution, containing an enzyme that digests lipids, changes over 10 minutes at two different temperatures.

The tables show the student's results.

Temperature at 37 °C

Time in minutes	1	2	3	4	5	6	7	8	9	10
pH	8.5	8.0	7.5	7.2	7.0	6.8	6.2	5.8	5.5	5.5

Temperature at 80 °C

Time in minutes	1	2	3	4	5	6	7	8	9	10
pH	9.2	9.0	8.8	8.8		8.6	8.6	8.4	8.4	8.4

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2. Question 4 · part b.i · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.i only.

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(i) Estimate a pH value for the lipid solution at 80 °C at 5 minutes.

(1)

(ii) Give the independent variable and the dependent variable for this investigation.

(2)

independent

dependent

(iii) Explain the differences in the results of the student's investigation at the two different temperatures.

(4)

(Total for Question 4 = 13 marks)

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2. Question 4 · part b.i · 1 mark · Working and answer

Working and answer

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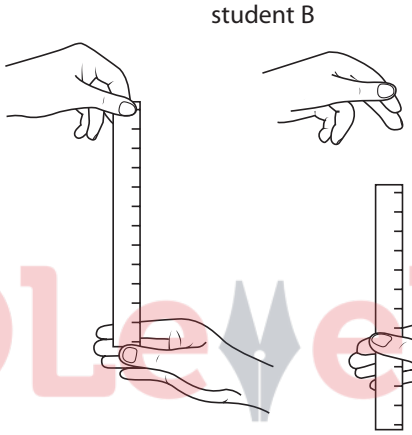
3. Question 5 · part e · 1 mark

November 2021 Paper 2

Further practice · Attempt part e only.

5 Two students use this method to investigate if caffeine affects the speed of a reflex action.

- student A sits in a chair holding out a hand with their thumb and forefinger open
- student B holds a metre rule with the 0 mm mark just above the open thumb and forefinger of student A
- student B releases the ruler
- student A catches the ruler as quickly as possible by closing their thumb and forefinger



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The method is repeated 10 times.

The student catching the ruler now drinks a cup of coffee and waits for 30 minutes.

The investigation is then repeated.

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3. Question 5 · part e · 1 mark

November 2021 Paper 2

Further practice · Attempt part e only.

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The table shows the student's results.

Conditions \ Trial	Distance travelled by ruler in mm										
	1	2	3	4	5	6	7	8	9	10	mean
without coffee	210	190	200	205	207	195	198	195	199	201	200
with coffee	165	160	163	167	165	170	175	170	180	175	

(a) Complete the table by calculating the missing mean value.

(2)

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(b) Coffee contains caffeine.

State the effect of caffeine on the reflex action.

(1)

(c) Explain why the student waited 30 minutes after drinking the coffee before continuing with the investigation.

(3)

3. Question 5 · part e · 1 mark

November 2021 Paper 2

Further practice · Attempt part e only.

(d) The student who catches the ruler does not eat or drink anything for several hours before the investigation.

Suggest why this improves the validity of the investigation.

(2)

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.....

(e) A cup of coffee contains 100 mg of caffeine. A can of cola contains 50 mg of caffeine.

If the student catching the ruler had drunk a can of cola instead of coffee, estimate the mean distance travelled by the ruler.

(1)

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mean distance = ... mm

(Total for Question 5 = 9 marks)

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Working and answer

4. Question 7 · part a.iii · 2 marks

June 2023 Paper 1

Further practice · Attempt part a.iii only.

7 A method that shows whether a person's kidneys are working correctly is to measure the volume of filtrate produced by the kidneys in one minute. This is known as the glomerular filtration rate (GFR).

A person with healthy kidneys has a GFR of over 100 cm^3 per minute.

The GFR of a person whose kidneys were not working correctly was measured once a year for nine years.

The table shows the results.

Year	2011	2012	2013	2014	2015	2016	2017	2018	2019
GFR in cm^3 per minute	80	75	79	79	67	60	50	45	35

(a) (i) Plot a line graph of the data on the grid.

(4)



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4. Question 7 · part a.iii · 2 marks

June 2023 Paper 1

Further practice · Attempt part a.iii only.

(ii) Explain why it is not possible to determine from the data when the person's kidneys stopped working correctly.

(2)

(iii) A person needs dialysis if the GFR falls to 15 cm^3 per minute.

Use the data to estimate when the person will probably need to start dialysis.

(2)

(iv) Discuss the accuracy of this estimate.

(3)

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4. Question 7 · part a.iii · 2 marks · Working and answer

Working and answer

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